



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105



Student Name: K. Archana

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Course Code: CSA0812

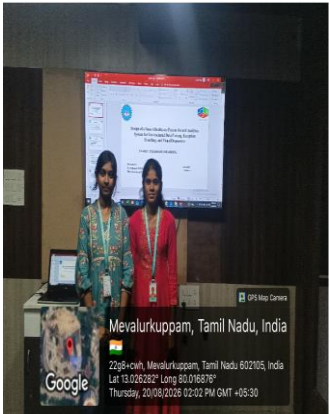
Slot: C

Course Name: Python Programming

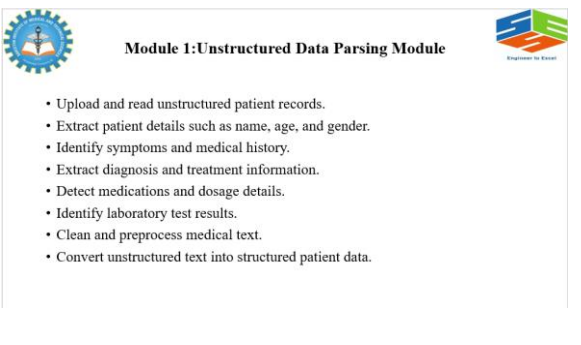
Course Faculty: Dr. Prabakaran J

Project Title: Design of a Smart Healthcare Patient Record Analytics System for Unstructured Data Parsing, Exception Handling, and Visual Diagnostics

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)

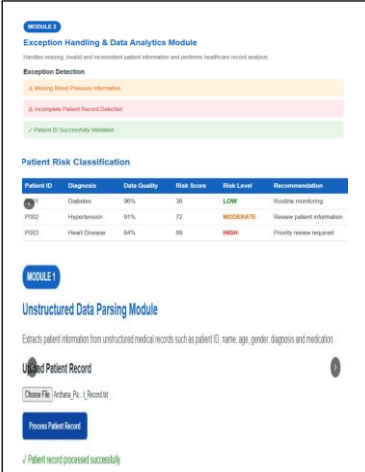


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Module 1: Unstructured Data Parsing Module

- Upload and read unstructured patient records.
- Extract patient details such as name, age, and gender.
- Identify symptoms and medical history.
- Extract diagnosis and treatment information.
- Detect medications and dosage details.
- Identify laboratory test results.
- Clean and preprocess medical text.
- Convert unstructured text into structured patient data.



MODULE 2
Exception Handling & Data Analytics Module

Handles missing, invalid and inconsistent patient information and performs healthcare record analysis.

Exception Detection

- ✓ Missing Blood Pressure Information
- ✓ Incomplete Patient Record Detected
- ✓ Patient ID Successfully Validated

Patient Risk Classification

Patient ID	Diagnosis	Data Quality	Risk Score	Risk Level	Recommendation
P001	Diabetes	95%	35	LOW	Routine monitoring
P002	Hypertension	91%	72	MODERATE	Review patient information
P003	Heart Disease	84%	88	HIGH	Priority review required

MODULE 1
Unstructured Data Parsing Module

Extracts patient information from unstructured medical records such as patient ID, name, age, gender, diagnosis and medication.

Upload Patient Record

Chennai File Archive Pa 1 Record

Process Patient Record

✓ Patient record processed successfully

Project Description: (here you write what you did in this project (contribution) including Model Description)

Module 1: The Unstructured Data Parsing Module extracts and organizes important patient information from unstructured medical records such as clinical notes, doctor reports, and uploaded patient files. It identifies details such as patient name, symptoms, diagnosis, medications, medical history, and test results, then converts them into a structured format for further analytics and visual diagnostics. The module uses text preprocessing, keyword extraction, and pattern-based parsing to identify relevant medical information from raw patient records. It cleans the input data, removes unnecessary information. This helps reduce manual data entry, improves data consistency, provides reliable input for the Exception Handling & Data Analytics Module and Visual Diagnostics Module.

Student Signature

Guide Signature